ASTANA IT UNIVERSITY

DIPLOMA PROJECT PRE-DEFENSE

OpenLesson

Development of a Web Platform for Planning, Monitoring, and Analyzing Faculty Open Classes in a Higher Education Institution

TEAM

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CLIENT

Astana IT University

Astana, October 2026

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One platform for the whole open-lesson cycle

Schedule → Materials → Poster → Attendance → Feedback → AI summary → Reports

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PART 1

Proposal, relevance, literature

Why AITU needs OpenLesson and what is already known

TASK 1

Project proposal

TASK 2

Evidence of relevance

TASK 3

Literature review

Task 1: Project proposal

PROJECT TITLE

Development of a Web Platform for Planning, Monitoring, and Analyzing Faculty Open Classes in a Higher Education Institution

Team: Dana Shokobalinova, Amina Sergazina, Nurdaulet Saparbekov
Supervisor: Yersultan Tulebayev
Client: Astana IT University

Problem

Open lessons are planned, announced, checked and reported by hand: no deadline control for slides and plans, feedback arrives late or gets lost, trimester reports are built manually.

Aim

Build OpenLesson, one web platform that automates planning, monitoring and analysis of open lessons, with AI-supported feedback summaries.

Objectives

1. Study the process: client + 4-5 teacher interviews
2. Review literature and similar systems
3. Requirements, system and UI design
4. Implement the 8 modules of the spec
5. Pilot one trimester and evaluate

Target users

- Administrator / methodologist
- Teacher giving the lesson
- Observing teacher
- Dean / deputy dean
- University leadership

Expected outcomes

- Working prototype with 8 modules
- Automated 10 / 7 / 3-day lesson cycle
- All feedback + AI summary in one cabinet
- Dashboard, Excel / PDF reports, docs

Project scope

In: 8 modules, 5 roles, responsive web, university e-mail, Excel import, AI as advice

Out: native app, video of lessons, full HR / LMS integration, AI decisions without a human

Task 2: Evidence of relevance

Meeting with the Academic-Methodological Department

Done

8 Oct 2026. The department requested OpenLesson and provided its specification.
Follow-up: Arailym Ilyassova answered our questions in six voice messages.

PARTICIPANTS

Shunkeyeva Saule Alisherovna, Head
Abisheva Sandugash Kanatovna, Chief manager
Arailym Ilyassova, Methodologist

1 teacher interview

Done

14 Oct 2026, 12 min, recorded with consent, transcribed. 4-5 interviews planned in total.

Respondent	Profile
B	Teacher at the School of Cybersecurity who has given open lessons

Respondent's name withheld. Quotes translated from Russian.

Hypothesis	Status	Evidence
H1 Schedule and forms are handled manually	**Confirmed**	Paper template, list via Google Form, e-mails sent by hand (B)
H2 Feedback arrives late and is generic	**Confirmed**	About 1 week via the dean, "everyone writes the same" (B)
H3 No summary analytics	**Indirect**	B asks for trimester statistics and a comparison across observers
H4 Staff will use a phone form	**Partly**	Fine for younger staff, harder for the older generation (B)

Task 2: Department answers

Arailym Ilyassova provided six voice replies to our follow-up questionnaire.
Written replies also clarified application fields, attendance and anonymity.

Topic	Department response	Status
Scheduling	7 schools. Schools approve dates with deans or deputies and send schedules to UMO. Lessons must avoid exams and holidays.	**Confirmed**
System users	Three UMO staff need access. Any faculty member may attend an open lesson, regardless of school.	**Confirmed**
Materials	No fixed plan or presentation template. Application fields follow the specification. File formats and limits remain to be agreed.	**Clarified**
Feedback	Anonymous feedback is not required. Who can see each review and its author still needs confirmation.	**Confirmed**
Changes	Proposed: UMO approves rescheduling requests. Teachers revise lesson materials after receiving UMO comments.	**Proposed**
Poster	Marketing should advise on the university template, colours and logo. UMO will coordinate the result.	**Follow-up**

Still to confirm: the three-day feedback rule, reminders, access rights, reporting and the minimum lesson quota period.

What the teacher told us → what OpenLesson must do

“The e-mail comes on the same day, at most a day before.”

→ Reminders 10, 7 and 3 days ahead (M2)

“Today they tell you, tomorrow you teach.”

Schools approve lesson dates with deans and send schedules to UMO (M1)

Feedback arrives after ~1 week via the dean and is posted in a group chat

→ Within 3 days, visible only to teacher, dean, methodologist (M6)

“No time to sit and write… everyone writes the same.”

→ Quick checklist of closed questions + one optional comment (M6)

“A recommendation from one person may be subjective.”

→ Compare several observers, AI summary of common points (M7, M8)

Phone is fine for me, harder for older colleagues

→ Simple UI that works on laptop and phone (non-functional)

Department clarification: feedback does not need anonymity. There is no fixed lesson plan template. Role access and the material submission rules still need confirmation.

Task 3: Literature review

13 sources reviewed

required ≥ 12, met

9 scientific articles, Scopus Q1-Q3

required ≥ 8, met

Search and selection

- Peer-reviewed journals and conference papers
- 2020-2026, teachers' professional development
- Free full text only
- Keywords: peer observation of teaching, online teacher professional development, MOOC, learning platform

Theme [sources]	What the literature agrees on	What is missing	Decision for OpenLesson
Peer observation of teaching [1-2]	Developmental, collegial observation is valued. Feedback needs strengths, weaknesses and comments	Digital support for the whole cycle. Only 3 of 13 observation tools are validated	**Status-based cycle, configurable rubric, teacher reply (M1-M3, M6)**
Online PD platforms and their design [3-8]	Dashboards and history support growth. Platform quality, usability and accessibility drive satisfaction. Moodle-type platforms suit university staff	Platforms train teachers, none runs open lessons and peer observation	**Teacher cabinet and dashboard, SUS at least 70, WCAG 2.1 AA (M8)**
Effect and uptake of online PD and MOOCs [9-13]	Online courses and MOOCs raise teachers' digital competences when they fit their needs. Usefulness, ease of use and time decide uptake	Acceptance data for peer-observation tools	**Reminders 10 / 7 / 3 days, UTAUT survey in the pilot (M2)**

Evidence base: 13 sources, 2020-2026: 2 reviews, 10 empirical and design studies, 1 conceptual paper. Full titles are on the References slide (at the end).

Research gap: every building block is studied separately. No source joins scheduling, deadline control, attendance, peer feedback and AI summaries for open lessons at a university. OpenLesson fills this gap.

PART 2

Comparison, technology, requirements

2

What already exists, which stack we chose and what the system must do

TASK 4

Existing systems

TASK 5

Technology

TASK 6

Requirements

Task 4: Existing system comparison

Criterion	Moodle + plugins	Google Calendar, Forms, Sheets	Evaluation (Education Advanced)	OpenLesson
Main purpose	Course management	Events and data collection	Teaching observation and evaluation	**The whole open-lesson cycle at AITU**
Scheduling	Calendar events	Google Calendar	Deadline reminders, shared class calendar not verified	**Approved trimester schedule from Excel, reminders 10 and 7 days ahead**
Visitor registration	Requires workflow configuration	Registration form	*Not verified*	**Invitation and AITU poster to all staff 3 days ahead**
Attendance	Additional Attendance plugin	Separate form or spreadsheet	*Not verified*	**Attendance sheet entered by the methodologist, QR optional**
Ratings and comments	Feedback questionnaires	Google Forms	Custom rubrics and comments	**8 criteria (1-5) + comments within 3 days, teacher replies**
Process monitoring	Adaptation to open classes	Statuses in Sheets	Evaluation completion tracking	**Status cycle, deadline control, automatic annulment**
Analytics and reports	Feedback and attendance reports	Spreadsheet analysis and export	Observation summaries and analytics	**Dashboard by year, school and department, Excel / PDF, AI summary**
University adaptation	Module and permission configuration	Connect tools and configure access	Configure forms, roles and workflows	**Built for AITU: 5 roles, Microsoft sign-in, staff from Digital University, KZ / RU / EN**

Not verified *= not confirmed in the reviewed documentation, not proof that the feature is missing. Evaluation mainly serves school organisations, so its fit for a university needs further assessment. Sources: Moodle docs (Calendar, Feedback, Attendance), Google Calendar and Forms help, Education Advanced Evaluation FAQ.*

Why a new platform: each tool covers one step. OpenLesson links schedule, materials, poster, attendance, feedback and reports to one lesson record, follows the AITU 10 / 7 / 3-day cycle and removes re-entering data. Time savings are measured in the pilot (Task 9).

Task 5: Technology comparison

Decision	Alternative	Main benefit	Consideration
Backend	Node.js and Express	Flexible JavaScript / TypeScript API	Module structure, validation and access checks are up to the team
	Node.js and NestJS	Modules, controllers, dependency injection	Framework conventions to learn
	Python and Django	Built-in ORM and admin interface	Server-rendered by default, REST needs an extra framework
	Python and FastAPI ✓	Typed validation (Pydantic), automatic OpenAPI docs, async	Structure of routers and services is up to the team
Database	**PostgreSQL ✓**	Relational integrity and SQL reporting	Design schema and migrations
	MySQL with InnoDB	Relational model and foreign keys	Team experience and hosting environment
	MongoDB	Flexible document structure	References, duplication and consistency to plan
API	**REST over HTTPS, JSON ✓**	Clear operations for lessons and feedback	Agree on endpoints and response formats
	GraphQL	Client selects fields and related data	Additional schema and resolvers
	WebSocket	Immediate notifications and updates	Persistent connections, added complexity

Sources: Express, NestJS, Django, FastAPI, PostgreSQL, MySQL and MongoDB documentation, MDN (REST, WebSocket), graphql.org. ✓ = chosen.

Chosen stack

Backend
Python, FastAPI, SQLAlchemy 2

Database
PostgreSQL (SQLite in development)

API
REST over HTTPS, JSON, OpenAPI docs

Frontend
React 19 + TypeScript, Vite, Tailwind

Why

- One language for the API, Excel import, PDF and posters, and the AI client
- A unique (lesson, teacher) pair blocks duplicate attendees and reviews. Foreign keys keep reports consistent
- REST covers every operation. WebSocket only if live updates become necessary

Task 6: Functional requirements

ID	Requirement	Module	Priority	Acceptance criterion
FR-01	**Sign-in and role-based access**	all	**Must**	Sign-in with the university Microsoft account, 5 roles. A teacher cannot change other teachers' lessons or reviews.
FR-02	**Trimester schedule**	M1	**Must**	Admin creates the academic year and imports the approved Excel schedule. Rows with missing required fields are reported by row number.
FR-03	**Edit, reschedule or annul a lesson**	M1	**Must**	Changes appear in the card and calendar. An annulled lesson keeps its reason in history and gets no poster or feedback forms.
FR-04	**Calendar, search and campus map**	M1	**Must**	Year, trimester, department and status filters, search by topic, course or teacher. The map highlights rooms with open lessons.
FR-05	**Reminders and notifications**	M2	**Must**	Reminders 10 and 7 days ahead in the app and by university e-mail. Sent history is kept and failed e-mails are retried.
FR-06	**Materials and deadline control**	M3	**Must**	Teacher uploads PPT / PPTX and a DOC / DOCX / PDF plan. At the deadline a lesson without both is annulled or flagged (policy).
FR-07	**Poster and invitations**	M4	**Must**	3 days ahead a poster with the AITU logo is generated for a confirmed lesson and sent to all active staff.
FR-08	**Record actual attendance**	M5	**Must**	The methodologist enters attendees from the attendance sheet. The same person cannot be added to a lesson twice.
FR-09	**Feedback form within 3 days**	M6	**Must**	Only confirmed attendees. Criteria on a 1-5 scale + comments, draft saved, a second submission is rejected.
FR-10	**Teacher reads feedback and replies**	M6	**Must**	Scores, averages, comments with authors, a reply and a PDF. The teacher cannot edit reviews of colleagues.
FR-11	**AI assistant**	M7	**Should**	Plan analysis, feedback summary and invitation text are shown as suggestions. A person makes every decision.
FR-12	**Dashboard and reports**	M8	**Must**	Statistics by year, trimester, school and department. Excel / PDF match the dashboard, annulled lessons are not counted as held.

Basis: *client specification (modules M1-M8), teacher interview and department replies (Task 2), literature (Task 3). Must = needed for the pilot, Should = desirable. QR check-in is an optional extra (spec 7.2).*

Task 6: Non-functional requirements and decisions

ID	Requirement	Priority	Acceptance criterion
NFR-01	Access security	Must	Every forbidden action in the role matrix is rejected, incl. direct API calls. 0 high-risk findings (OWASP ZAP).
NFR-02	Data integrity	Must	Concurrent operations create no duplicate attendees or reviews. A failed operation leaves no partial records.
NFR-03	Response time	Must	p95 < 500 ms for list and lesson pages with 100 concurrent users on staging.
NFR-04	Responsive interface	Must	Core actions work at 360, 768 and 1440 px without overlapping fields or buttons.
NFR-05	Usability	Should	SUS ≥ 70, ≥ 90 % task success in the 4 key scenarios, including older staff.
NFR-06	Accessibility	Must	WCAG 2.1 AA: 0 critical issues in axe, full keyboard walkthrough.
NFR-07	Languages	Should	Kazakh, Russian and English interface, e-mails in the user's language.
NFR-08	Reliability and recovery	Should	Failed e-mails are retried. A backup restores with matching record counts and valid relations.

Targets are checked in Task 9 (load test, usability test, axe, ZAP).



Open questions: status

Who confirms attendance

Confirmed

The Academic-Methodological Department marks the paper sheet, then the methodologist enters it.

Criteria and scale

Proposed

8 criteria on a 1-5 scale. The admin can edit the list. Checked in the usability test.

Who sees comments

Open

Spec: the teacher, dean and methodologist only. The department still has to confirm who sees each review and its author.

Anonymous feedback

Confirmed

Not required, the department confirmed. Reviews stay named, as in the spec.

Editing a sent review

Spec

Not allowed. The form stays a draft until it is sent once.

PART 3

Architecture, UX, evaluation

How OpenLesson is built, how people use it and how we prove it works

3

TASK 7

Architecture

TASK 8

UX / prototype

TASK 9

Verification

Task 7: Architecture and diagrams

ARCHITECTURE STYLE

Modular monolith: React single-page app → FastAPI REST API → relational database, with a background automation cycle and Microsoft, AI and Digital University integrations

Client

React 19 SPA (TypeScript, Tailwind) for laptop and phone, role-based screens for 5 roles in Kazakh, Russian and English

REST API

FastAPI, JSON over HTTPS, OpenAPI docs, access checked per role and per school (dean)

Automation cycle

APScheduler every 5 min: reminders at 10 / 7 days, deadline check, poster at 3 days, 3-day feedback window

Data

PostgreSQL (SQLite in development) via SQLAlchemy, file storage for slides, plans and posters

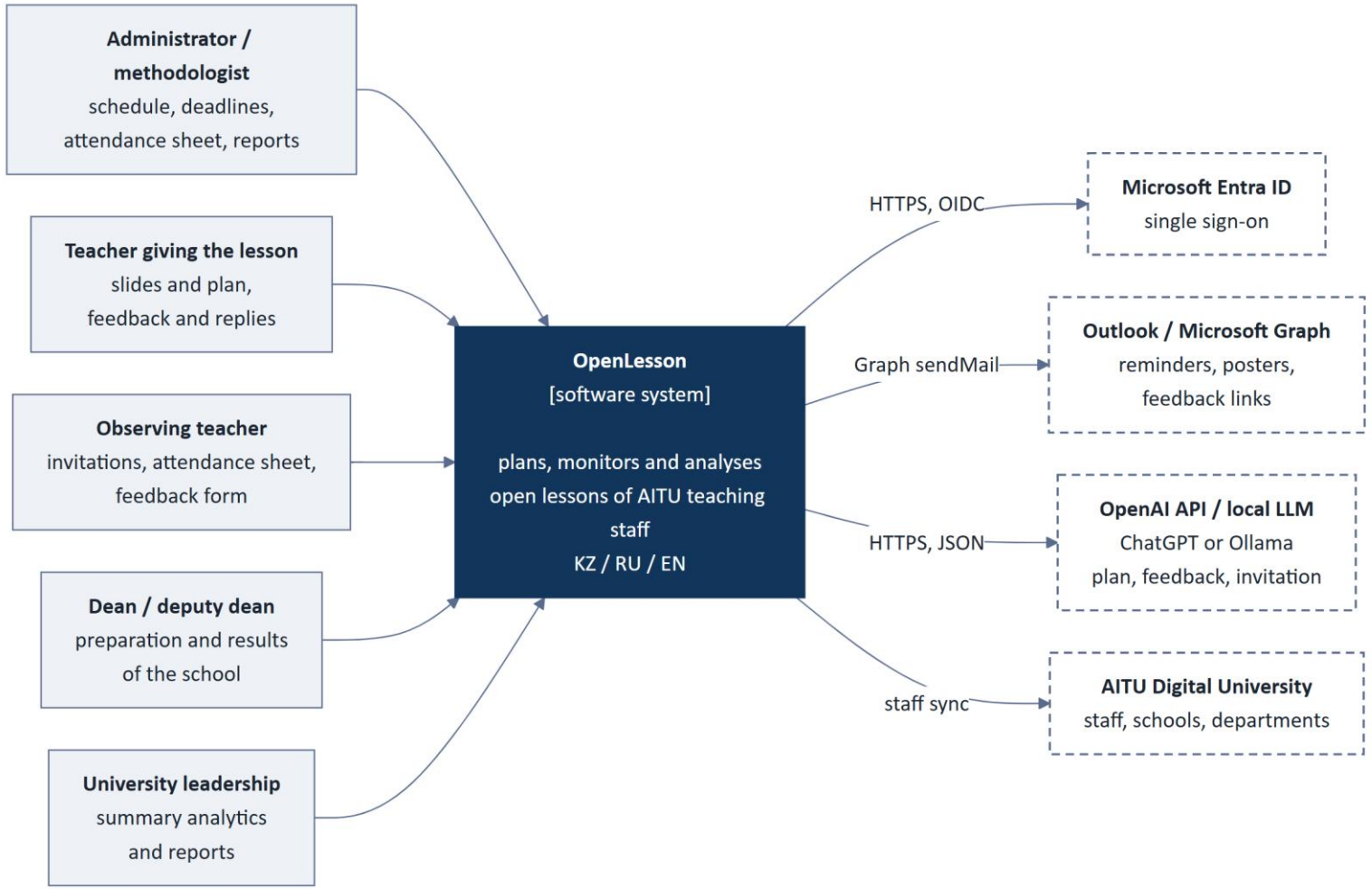
Integrations

Entra ID (OpenID Connect), Outlook via Graph, OpenAI API or a local LLM for AI advice, AITU Digital University

Reliability

E-mail outbox with retries, audit history per lesson, idempotent cycle, AI with local fallback

System context diagram



Mermaid source: docs/diagrams/01-system-context.mmd

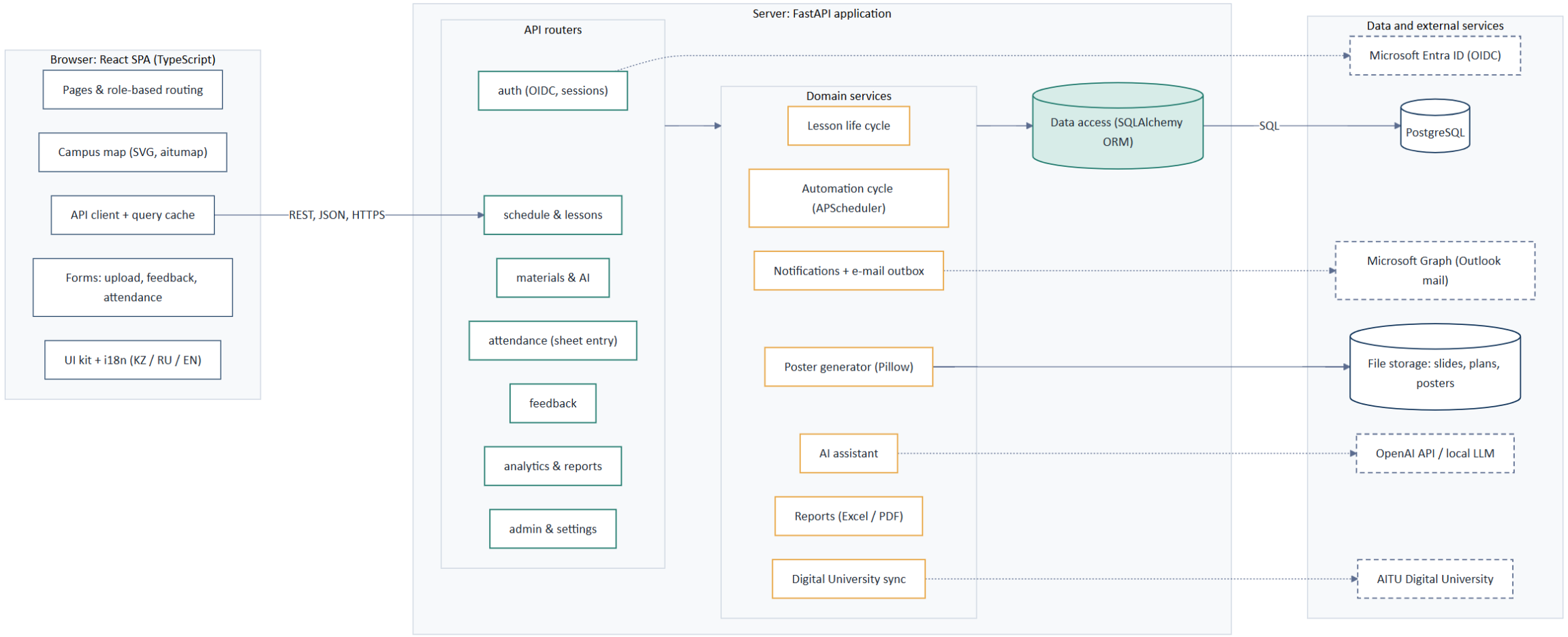
Use case diagram



orange = started by the scheduler (time trigger)

Mermaid source: docs/diagrams/02-use-cases.mmd

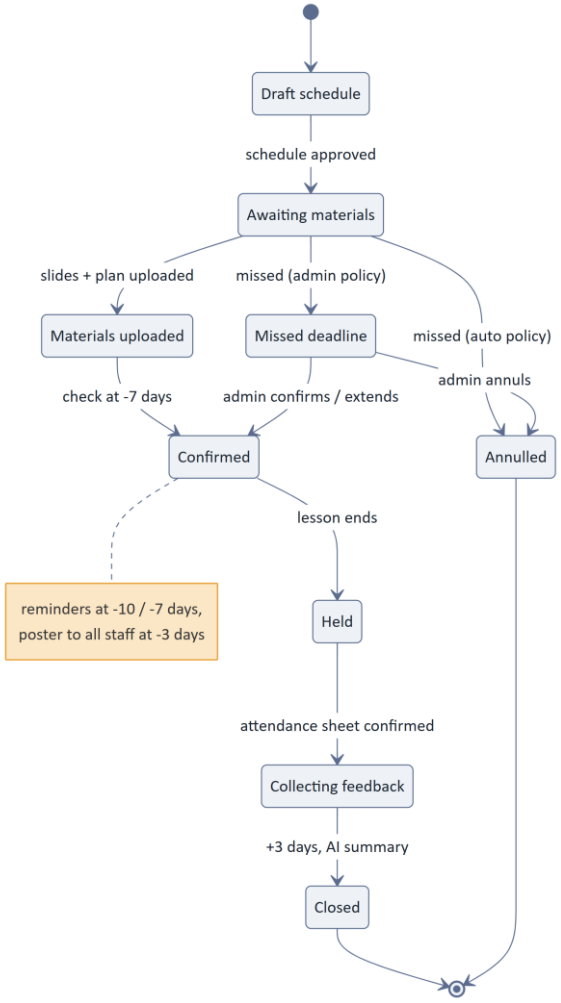
Component diagram (initial draft)

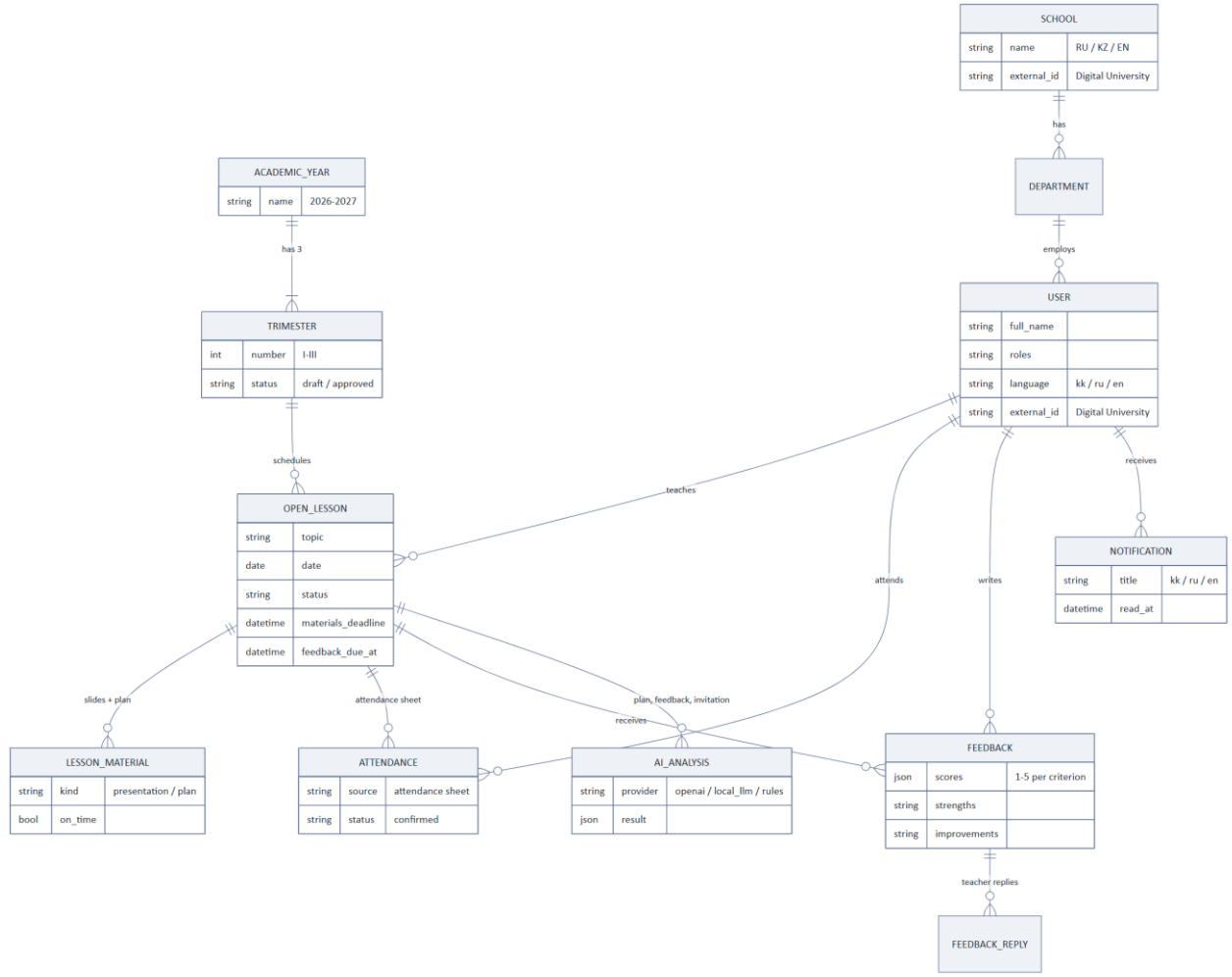


dashed = external service

Mermaid source: docs/diagrams/03-components.mmd

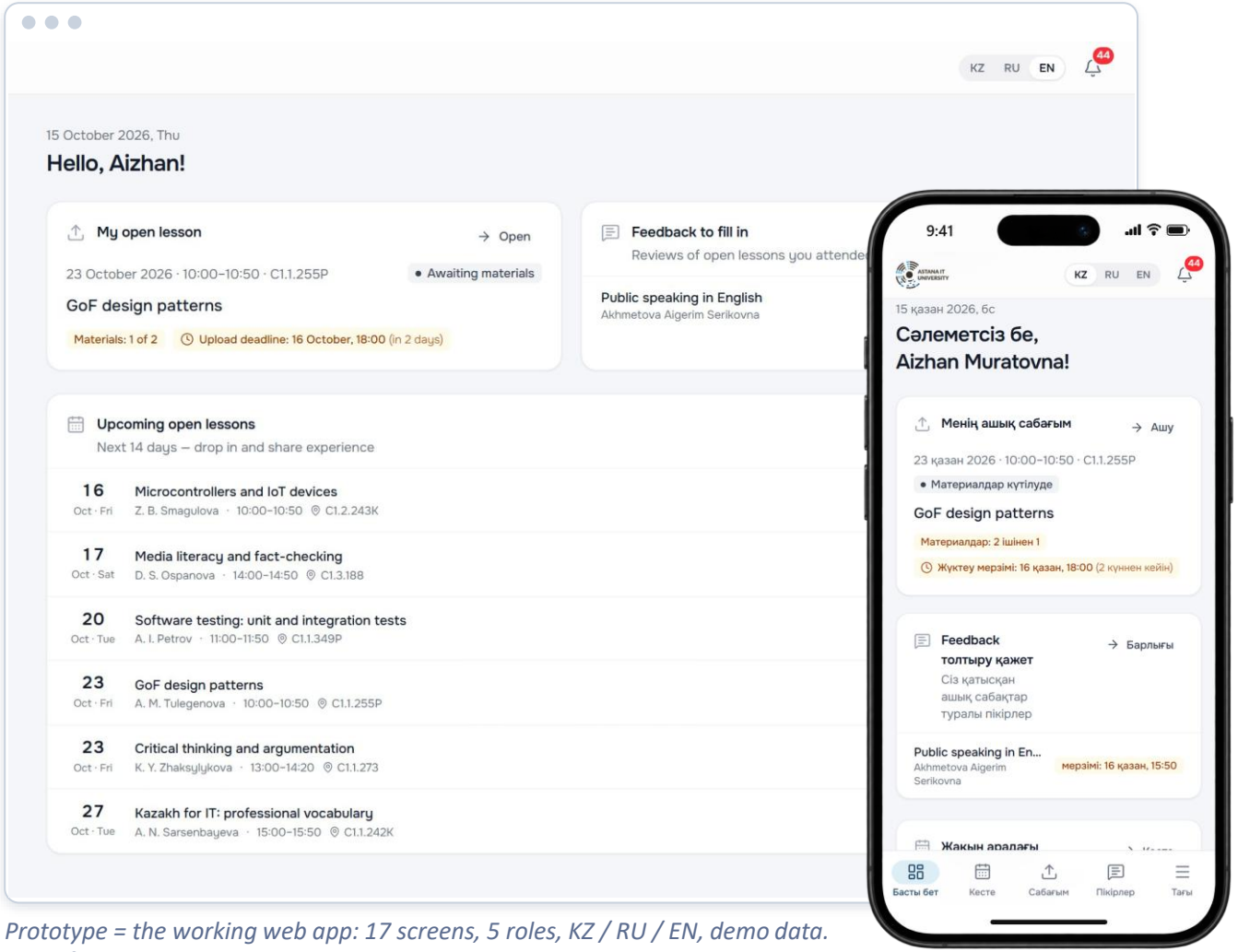
Core data model and lesson life cycle





Mermaid source: docs/diagrams/04-lesson-lifecycle.mmd, docs/diagrams/05-data-model.mmd

Task 8: UX / prototype



Prototype = the working web app: 17 screens, 5 roles, KZ / RU / EN, demo data.
Next: four key scenarios on a phone.

Phone first

Every screen works at 390 px. A bottom tab bar keeps the four main sections of each role one tap away

Deadlines always visible

Countdown chips, “1 of 2” progress, reminders 10 and 7 days ahead instead of a same-day e-mail

Feedback in two minutes

8 criteria × one big tap target (1-5), two short text fields and an autosaved draft for staff with “no time to sit and write”

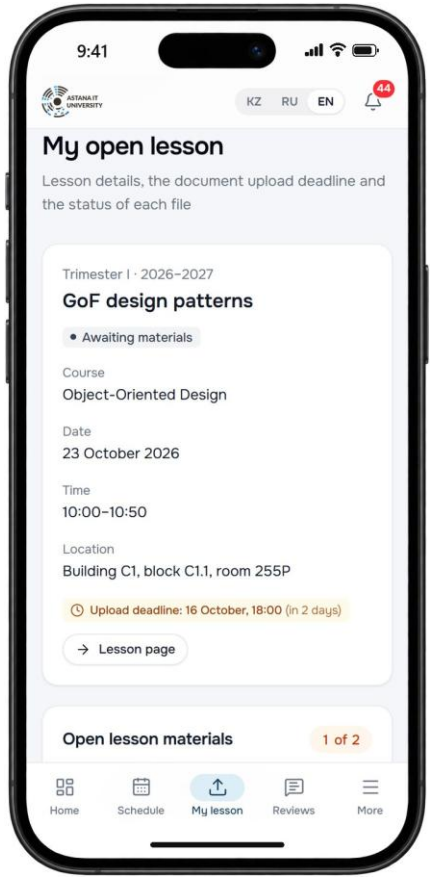
Accessible by default

WCAG 2.1 AA: keyboard focus, contrast, labels for screen readers, reduced motion

Kazakh, Russian, English

KZ / RU / EN switch in the header. Notifications, e-mails and posters follow the user's language

Scenario 1: Teacher prepares the lesson and reads feedback

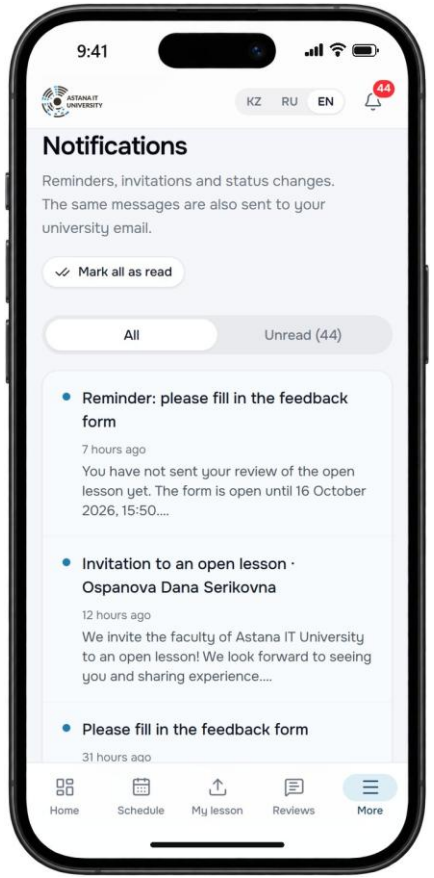


1

Upload slides and plan

"My open lesson": date, room, deadline and "1 of 2" progress. Upload PPTX / DOCX / PDF from the phone. The AI assistant analyses the plan right away.

Design choice
a checklist with progress, so the teacher always sees what is still missing.

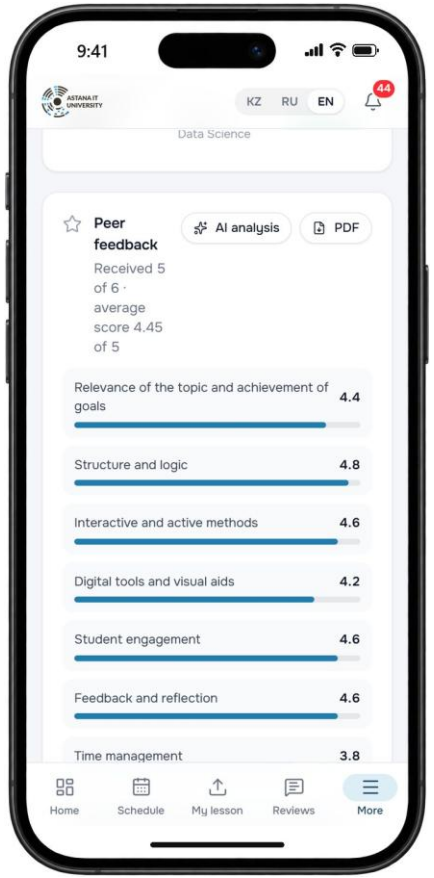


2

Get reminded in time

Reminders 10 and 7 days before the upload deadline, invitations to colleagues' lessons and feedback requests, both in the app and by e-mail.

Design choice
the same message in two channels. The bell shows the unread count.



3

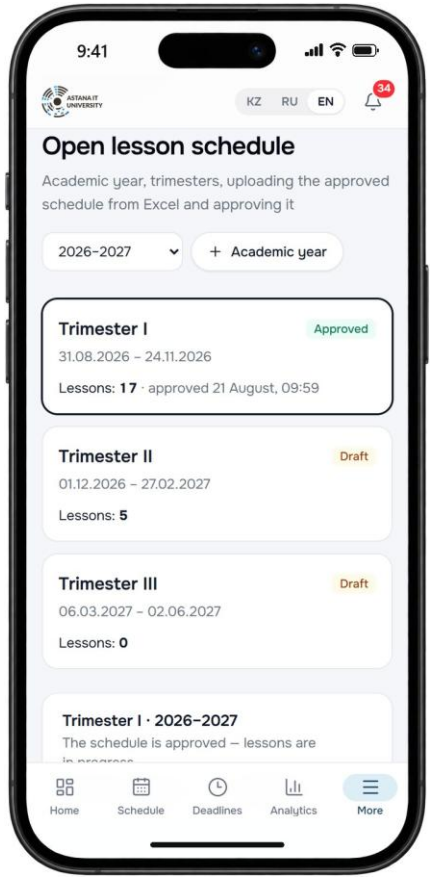
Read feedback and reply

Average per criterion, an AI summary of strengths and improvements, every review with the colleague's name and a reply under it, PDF export.

Design choice
overview first (scores, summary), individual reviews below.

Teacher's bottom bar: Home, Schedule, My lesson, Reviews, More

Scenario 2: Methodologist runs the trimester

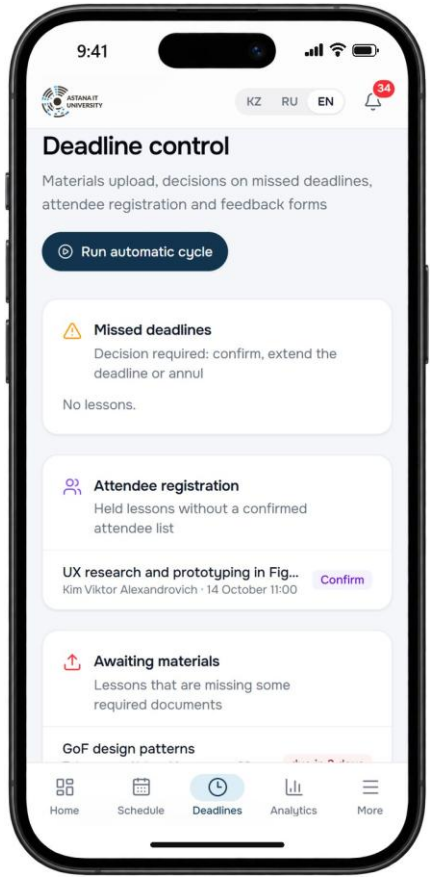


1

Import and approve the schedule

The approved Excel schedule per trimester. Wrong rows are reported by row number. Approval creates the lessons and notifies every teacher.

Design choice
the schedule stays a draft until approved, so mistakes are fixed before anyone is notified.

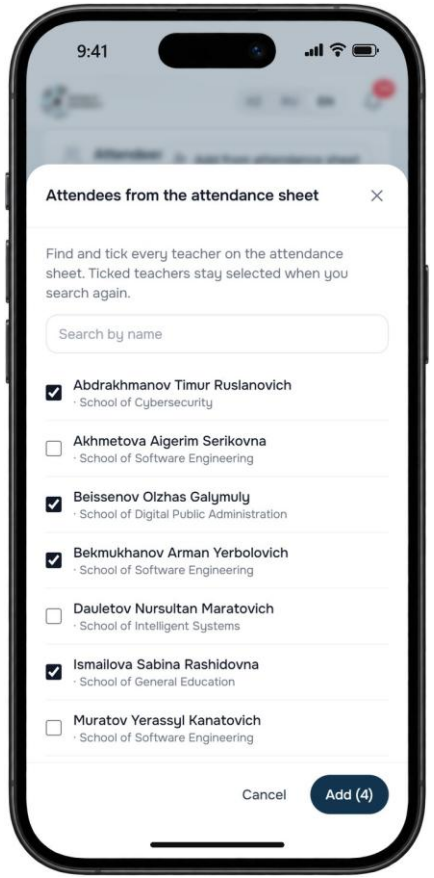


2

Control deadlines

Missed deadlines that need a decision, attendee registration, lessons without materials sorted by deadline, unsent feedback forms.

Design choice
an inbox of exceptions: only what needs action, most urgent first.



3

Enter the attendance sheet

After the lesson the methodologist ticks everyone from the paper attendance sheet and confirms. Feedback forms then open for all attendees at once.

Design choice
multi-select search keeps ticked names. On a phone it opens as a bottom sheet.

Methodologist's bottom bar: Home, Schedule, Deadlines, Analytics, More

Scenario 3: Colleague is invited, attends and gives feedback



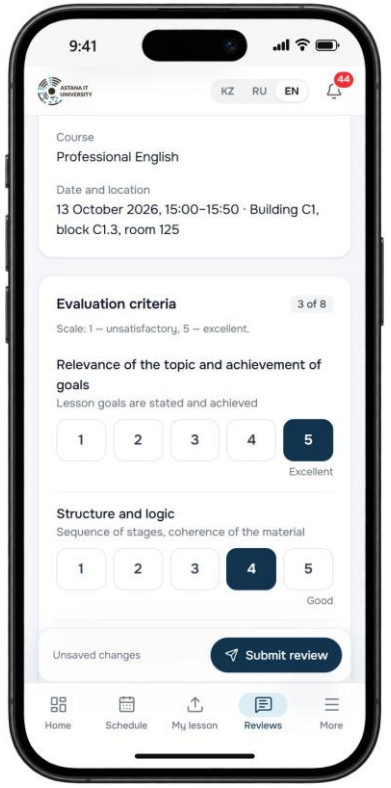


ATTENDANCE SHEET

Open lesson, 15 Oct, 10:00, room C1.2.243K

№	Full name / position	✓
1	*Abdrakhmanov T. R.*	✓
2	*Beissenov O. G.*	✓
3	*Bekmukhanov A. Y.*	✓
4	*Ismailova S. R.*	✓
5	*Kim V. A.*	✓

Marked by: Academic-Methodological Department



1 Gets the invitation

Three days before the lesson every teacher receives the generated AITU poster by e-mail and in the app.

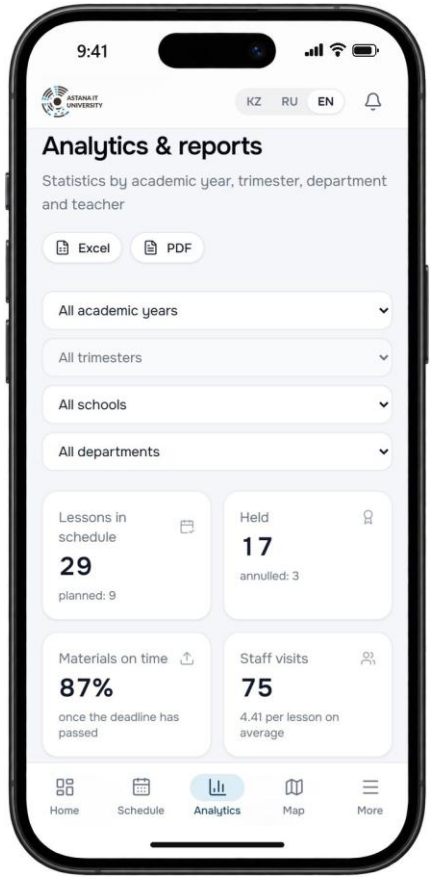
2 Is marked on the sheet

A staff member of the Academic-Methodological Department marks who attended. The methodologist then enters the list.

3 Fills in feedback

8 criteria with big tap targets, autosaved draft, one confirmation. Reminders after 1 and 2 days. The 3-day window closes by itself.

Scenario 4: Dean and leadership monitor results

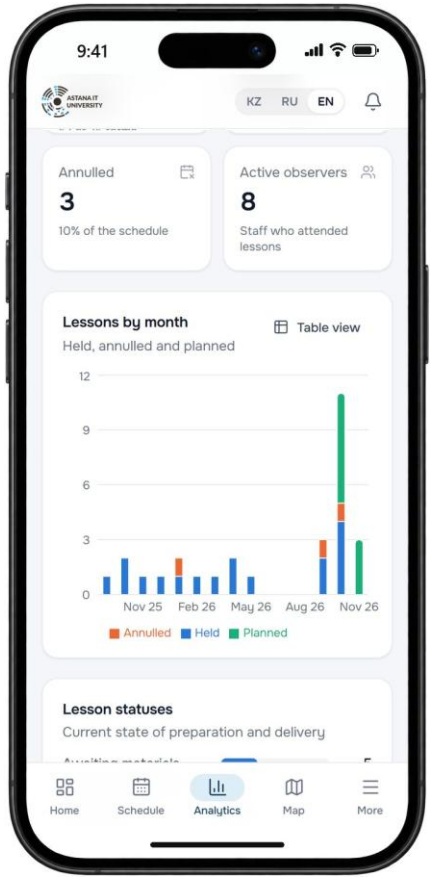


1

Filter and compare

Year, trimester, school and department filters. Held and annulled lessons, materials on time, visits, feedback rate, average score.

Design choice
a dean automatically sees only their own school.

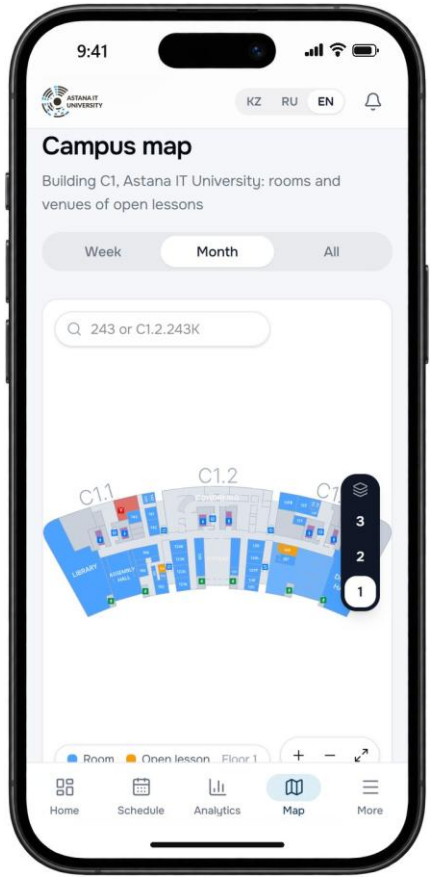


2

Charts with a table view

Lessons by month, statuses and average score per criterion. Every chart switches to a table, Excel / PDF export in one tap.

Design choice
the same status colours as in the schedule.



3

Find the room

The campus map of building C1 (aitumap) highlights rooms with open lessons. Search by room number.

Design choice
touch-friendly map: pinch to zoom, floor switcher.

Leadership's bottom bar: Home, Schedule, Analytics, Map, More

Task 9: Verification and evaluation plan

1. Automated tests

Unit and API integration tests on an isolated database: life cycle, deadline policies, permissions, attendance rules, imports, reports.

pytest, FastAPI TestClient, demo clock

18 tests pass

2. Scenario tests

The four Task 8 scenarios run end-to-end in a real browser for every role, on a laptop and a phone, in all three languages.

Playwright, seeded demo data

planned for Nov 2026

3. Quality gates

Performance, accessibility and security are checked on staging before real users see the system.

Locust, Lighthouse / axe, OWASP ZAP

planned for Nov 2026

4. User evaluation

Moderated usability test, then a pilot trimester at one school compared with the current manual process.

think-aloud, SUS, logs, survey

planned for Nov 2026 - Feb 2027

Already verified on the working prototype

18

automated tests pass (pytest), incl. KZ / RU / EN messages

34

demo lessons taken through the full life cycle by the scheduler

75

feedback forms filled in and summarised (AI + rule-based fallback)

1,095

staff records loaded (export, later AITU Digital University), re-import idempotent

Evaluation metrics: baseline vs target

Metric	Baseline: current manual process	Target with OpenLesson	How it is measured
Time until feedback reaches the teacher	≈ 1 week, via the dean (interview B)	**≤ 3 days for ≥ 90 % of lessons**	lesson end → feedback window closed (event log)
Reminder lead time	same day, at most 1 day before (B)	**10, 7 and 3 days ahead (100 % sent)**	e-mail outbox and notification log
Feedback completion	To be measured in the pilot baseline	**≥ 80 % of confirmed attendees**	submitted forms ÷ confirmed attendances
Materials on time	not tracked	**≥ 90 % of lessons**	upload time vs deadline in the event log
Time to fill in one review	paper template, timed in the usability test	**median ≤ 3 min**	form opened → submitted (timestamps)
Methodologist effort per trimester	hours from a time log in trimester I	**-50 %**	the same time log in trimester II

Baseline sources: Task 2 interview (B) and records of trimester I 2026-27, which still runs the old way (e-mails, Google Form, paper template). The same school and people are measured again in trimester II with OpenLesson.

Product quality targets

≥ 70

Usability, SUS score

10-item SUS questionnaire per role after the usability test and after the pilot (68 = average across studies).

When: Nov 2026 and Feb 2027

≥ 90 %

Task success, 4 key scenarios

Moderated think-aloud test with 5-8 users, including older staff (H4), time on task recorded.

When: Nov 2026

< 500 ms

API response time, p95

Locust load test on staging: 100 concurrent users ≈ 10 % of the 1,095 staff, mixed read / write.

When: Nov 2026, before the pilot

0

Critical accessibility issues

axe and Lighthouse (score ≥ 90) on all 17 screens plus a keyboard-only walkthrough (WCAG 2.1 AA).

When: every release

0

High-risk security findings

OWASP ZAP baseline scan and an automated role-access matrix that checks every endpoint for all 5 roles.

When: every release

≥ 4 / 5

AI summary usefulness

Teachers rate each summary. The methodologist checks 20 summaries against the reviews: 0 invented points.

When: during the pilot

Quality gates: the pilot starts only when the performance, accessibility and security targets are met on staging.

Measurement conditions and schedule

Environment
Staging with the production setup: PostgreSQL, HTTPS, Microsoft sign-in, staff data from AITU Digital University. Automated tests: isolated SQLite and a demo clock that runs a trimester in minutes.

Participants
One pilot school: about 20-30 teachers, its methodologist and dean. Usability test: 5-8 people of different ages and roles, in Kazakh, Russian and English.

Baseline
Trimester I 2026-27 runs the old way: dates from e-mails and Google Form, paper reviews, the methodologist's time log, a short survey.

Comparison
Same school, same people, same metrics in trimester II. Report medians and ranges. With a small sample we do not claim statistical significance.

Threats
Novelty effect, small sample, a different mix of lessons → full-trimester logs and follow-up with teacher B and pilot participants.

Schedule

24 Oct 2026
Pre-defense. Baseline collection starts in trimester I

Oct-Nov 2026
Scenario tests and quality gates on staging, fix findings

Nov 2026
Usability test: 5-8 users, SUS, task success, time on task

Dec 2026 - Feb 2027
Pilot in trimester II at one school with real open lessons

Mar 2027
Compare with the baseline and write the evaluation chapter of the thesis

Dependency: the pilot needs AITU IT to register the app in Microsoft Entra ID (sign-in and Mail.Send).

Success criterion: all quality gates pass, and at least 4 of the 6 process metrics reach their target in the pilot trimester.

Summary and next steps

Done

- Problem confirmed: client spec, 1 teacher interview, 13 sources
- 12 functional and 8 non-functional requirements
- Working prototype: 8 modules, 5 roles, 17 screens, KZ / RU / EN
- 18 automated tests pass, 1,095 staff records loaded
- GitHub and YouTrack: every task has an owner, a branch and a reviewed pull request

Next

- Oct-Nov 2026: scenario tests and quality gates on staging
- Nov 2026: usability test with 5-8 users (SUS, task success)
- Dec 2026 - Feb 2027: pilot in trimester II at one school
- Mar 2027: compare with the trimester I baseline

Needed from AITU

- Register the app in Microsoft Entra ID (sign-in, mail)
- Access to AITU Digital University staff data
- A pilot school with its methodologist and dean
- Confirm who sees each review and its author

OpenLesson joins scheduling, deadline control, attendance, peer feedback and AI summaries for open lessons in one system. This is the gap found in the literature review.

Project Management and Risks

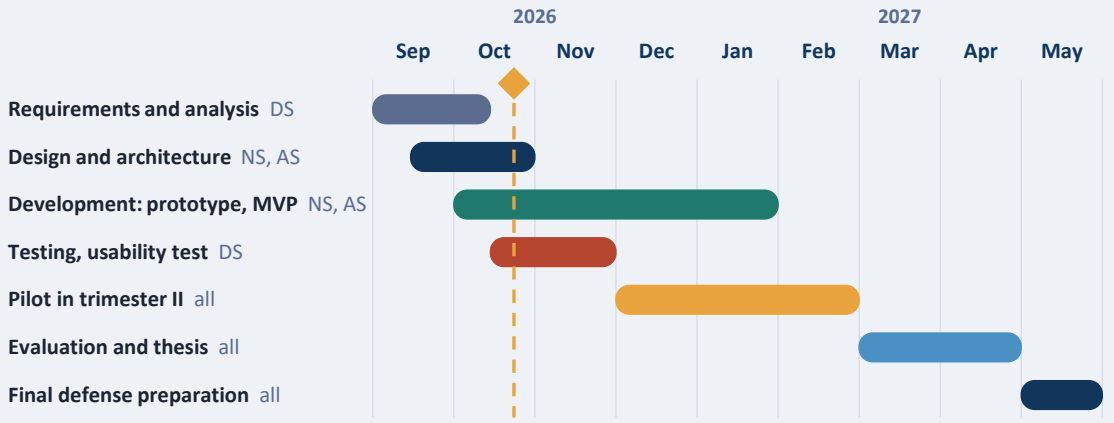
1 Team responsibilities

DS — Dana Shokobalinova
PM, research, QA
Plan and YouTrack, contact with supervisor and department, interviews, testing and evaluation

NS — Nurdaulet Saparbekov
Backend and architecture
FastAPI and database design, automation cycle, integrations: Entra ID, Outlook, Digital University, AI

AS — Amina Sergazina
Frontend and UI / UX
React interface and mobile version, UX prototype and usability, KZ / RU / EN localisation

2 Project timeline



3 GitHub and YouTrack

GitHub

code, docs, pull requests

- Branch per issue: DIP-20-excel-import
- Conventional Commits: feat, fix, docs, test
- Pull request reviewed by a teammate
- Tag prob-period-1 on the submitted version

```
feat(schedule): import Excel schedule
Refs: DIP-20
```

YouTrack, project DIP

42 tasks, 5 milestones, 5 risks

Backlog | Develop | Review | Done

DIP-4	README and team rules	DS
DIP-20	M1: schedule and Excel import	NS
DIP-33	Teacher pages	AS
DIP-39	Automated backend tests	DS

4 Project risks

Risk	Probability	Impact	Mitigation
No access to the AITU Microsoft tenant in time	Medium	High	Demo sign-in and console e-mail until AITU IT registers the app, request sent early
Older staff find the system hard to use	Medium	High	Phone-first simple UI in 3 languages, usability test with older staff
Requirements change (anonymous feedback, lesson plan)	Medium	Medium	Settings and criteria are configurable, regular check-ins with the department
Personal data sent to an external AI service	Low	High	Local LLM option, only review text is sent, AI only advises
Delays because of study load	Medium	Medium	2-week iterations on the YouTrack board, MoSCoW scope, buffer in April

Supervisor Feedback and Improvements

#	Supervisor's feedback	Action taken	Status
1	**Task 2: show the meeting with the Academic-Methodological Department** Name the participants and say that the project is the department's request.	• Added the 8 Oct 2026 meeting with S. A. Shunkeyeva (Head) and S. K. Abisheva (Chief manager) • Stated that OpenLesson and its specification come from the department	✓ Completed
2	**Literature review: full titles, one slide** Give the full title of every publication and merge the review into one slide.	• Added a References slide (at the end): 13 entries in APA 7 with DOI • 6 review slides merged into one: themes, gaps, decisions, research gap	✓ Completed
3	**Add Project Management and Risks** Team roles, timeline from September 2026 to May 2027, GitHub, YouTrack, risks.	• Added roles of the 3 members, a timeline up to the May 2027 defense and a risk register • Each risk has a probability, an impact and a mitigation	✓ Completed
4	**Run the project in GitHub and YouTrack** Show progressive commits and tasks, and what each student does.	• Public repository: every task has its own branch and a pull request reviewed by another member • YouTrack: due dates, 5 milestones, 65 dependencies and 5 risk issues (next slide)	✓ Completed

Example of an implemented improvement: the literature review

Before

6 slides with short citations like "Grech (2024), Malta"



After

1 review slide and a full reference list



Task 3: Literature review



References

GitHub and YouTrack in Practice

3 members, each commits from their own account

47 YouTrack issues: 42 tasks and 5 risks

5 milestones up to the May 2027 defense

65 dependency links between tasks

30+ pull requests, each reviewed by a teammate

1 GitHub

github.com/Garryflop/OpenLesson

- **Public repository**
  the supervisor and the committee see the full history
- **README**
  problem, users, outcomes, team, scope, status, stack, links, setup and tests
- **Documentation in docs/**
  proposal, literature review, existing systems, technologies, requirements, diagrams, UX prototype, evaluation plan
- **Own accounts**
  every member commits under their own name: Conventional Commits with Refs: DIP-N
- **Branch and pull request per task**
  reviewed by another member and merged into a protected main
- **Tests and CI**
  backend tests (pytest) and the frontend build run on every pull request

2 YouTrack

zarkinzz.youtrack.cloud, project DIP

- **Project DIP**
  42 tasks over research, requirements, design, prototype, implementation and testing
- **Every task**
  description, acceptance criteria, assignee, priority, due date and status
- **Milestones**
  pre-defense 24.10.2026, MVP 30.11.2026, pilot 28.02.2027, thesis 30.04.2027, defense May 2027
- **Dependencies and risks**
  65 links between tasks and 5 risk issues with owners (DIP-44 to DIP-48)
- **Linked to GitHub**
  commits and pull requests appear in their task through the branch name and Refs
- **Board**
  columns Backlog, Develop, Review, Done. A task is Done after its acceptance criteria are checked

Code layout: backend/app and frontend/src, tests in backend/tests, configuration in .github/ and .env.example. Top-level diagrams/ and prototype/ folders, as the committee recommended, follow after the pre-defense.

Materials and Links

Repository, version prob-period-1: https://github.com/Garryflop/OpenLesson/tree/prob-period-1/
YouTrack: zarkinzz.youtrack.cloud/issues/DIP **Run the prototype:** README, Setup **Presentation sources:** docs/presentation

#	Section	In the repository (all links open the version tagged prob-period-1)
1	**Project proposal**	docs/proposal.md
2	**Evidence of relevance**	docs/relevance.md
3	**Literature review**	docs/literature-review.md
4	**Existing system comparison**	docs/existing-systems.md
5	**Technology comparison**	docs/technology-comparison.md
6	**Requirements specification**	docs/requirements.md
7	**Architecture and diagrams**	docs/diagrams: editable Mermaid sources (.mmd) and rendered SVG
8	**UX / prototype**	docs/prototype.md (screens), prototype code: backend, frontend
9	**Verification and evaluation**	docs/evaluation-plan.md, automated tests: backend/tests
10	**Project management and risks**	docs/project-management/plan.md, tasks and risks in YouTrack
11	**Supervisor feedback and improvements**	docs/supervisor-feedback.md

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Thank you!

Questions and comments are welcome

OpenLesson: planning, monitoring and analysis of open lessons at Astana IT University